# 1 Unweighted data selection for linear regression

Hanneke, Moran, Shlimovich, and Yehudayoff [2] ask, in Question 3 of their COLT 2025 open-problem paper, for the worst-case cost of *unweighted* data selection for vector-valued linear regression under the minimum-Frobenius-norm empirical risk minimizer (ERM). In this note, we give a complete answer for this problem.

## 1.1 Setting and main result

Let $d \geq 1$ be input dimension, $m \geq 1$ be output dimension, $n$ be selection budget and $N \gg n$ be the size of the full dataset. Let $\mathcal{Z} = \mathbb{R}^d \times \mathbb{R}^m$. A linear predictor is a matrix $W \in \mathbb{R}^{m \times d}$, acting as $x \mapsto Wx$, with squared loss $\ell_z(W) = \|Wx - y\|_2^2$ for $z = (x, y) \in \mathcal{Z}$. For a finite dataset $D = \{z_i\}_{i=1}^N \subseteq \mathcal{Z}$, define its average loss $L_D(W)$ and minimum achievable loss $L_D^\star$ as

$$L_D(W) = \frac{1}{N} \sum_{i=1}^{N} \ell_{z_i}(W), \qquad L_D^\star = \min_W L_D(W).$$

**The learning rule.** We consider min-norm ERM. In particular, given a finite nonnegative combination $\sum_j c_j \, \ell_{z_j}(W)$ ($c_j > 0$), the minimizer set is a nonempty affine subspace. The learning rule $\mathcal{A}$ returns the minimizer with smallest Frobenius norm, this is the *min-norm ERM* of [2].

**Unweighted selection.** For a budget $n$, define $L_D^\star(n)$ to be the minimum dataset loss obtained on an unweighted subset of size $n$,

$$L_D^\star(n) \;=\; \min_{z_1, \ldots, z_n \in D} \; L_D\left(\mathcal{A}\left(\tfrac{1}{n} \sum_{i=1}^{n} \ell_{z_i}\right)\right),$$

and $\mathcal{F}(d, m, n)$ be the supremum ratio between the best full-dataset loss achievable by training on $n$ selected examples—using the uniform average of the selected losses—and the optimal full-dataset loss.[1]

$$\mathcal{F}(d, m, n) \;=\; \sup_D \frac{L_D^\star(n)}{L_D^\star},$$

**Theorem 1.1** (Answer to Question 3 of [2] for $d \geq 1$)**.** *For all integers $d \geq 1$, $m \geq 1$, $n \geq 1$,*

$$\mathcal{F}(d, m, n) \;=\; \begin{cases} d + 1, & n \geq d, \\ \infty, & n < d. \end{cases}$$

**Remark 1.2** (Comparison with weighted data selection for regression)**.** For *weighted* selection—in which the selected losses may be combined with arbitrary nonnegative weights rather than uniformly averaged—Hanneke, Moran, Shlimovich, and Yehudayoff established a sharp trichotomy [2, 1] for the analogous worst-case ratio $\mathcal{F}^{\mathrm{w}}$:

$$\mathcal{F}^{\mathrm{w}}(d, m, n) \;=\; \begin{cases} 1, & n \geq 2d, \\ d + 1, & n = d, \\ \infty, & n < d, \end{cases}$$

[1] For $n \geq d$ our upper bound shows $L_D^\star = 0$ implies $L_D^\star(n) = 0$, so excluding the degenerate ratios $0/0$ is immaterial; for $n < d$ we prove the lower bound with instances satisfying $L_D^\star > 0$, so the answer is also unaffected by how one interprets ratios with vanishing denominator.

The unweighted version exhibits a sharp contrast with weighted version, since adding data for $n \geq d$ does not help in the worst case.

**Conventions.** Throughout, $d, m \geq 1$ and $n \geq 1$ are integers. Following the notation $\min_{z_1,\dots,z_n \in D}$ of [2], a *selection* is a tuple $(z_1, \dots, z_n) \in D^n$; repetitions are allowed. (The lower bounds hold verbatim for selections required to be distinct; the upper bound uses repetitions only to fill the exact budget, and is also valid under an "at most $n$ examples" convention without repetitions.)

### 1.2 Proof of Theorem 1.1

The case $n < d$ and the upper bound for case $n \geq d$ is already derived in [1]. The remain goal is to prove $\mathcal{F}(d, m, n) \geq d + 1$ when $n \geq d$. It suffices to give a lower bound when $m = 1$; predictors are vectors $w \in \mathbb{R}^d$ and the rule is the min-Euclidean-norm ERM $\mathcal{A}^\star$.

We first review the lower bound instance for $n = d$. Fix $d \geq 1$ and set

$$c > \sqrt{2}\,(d+1), \qquad S_d := \sum_{i=1}^{d} i = \frac{d(d+1)}{2}.$$

Define the $(d+1)$-point scalar dataset $P = \{p_0, p_1, \dots, p_d\}$, where

$$p_i = (e_i,\ ic) \quad (i \in [d]), \qquad p_0 = (x_0, y_0), \quad x_0 = -\sum_{i=1}^{d} e_i, \quad y_0 = -cS_d - (d+1).$$

Write $\Phi_P(w) = \sum_{i=1}^{d}(w_i - ic)^2 + (\langle w, x_0\rangle - y_0)^2$ for the unnormalized loss on $P$, and note the convenient form of the residual on $p_0$:

$$\langle w, x_0\rangle - y_0 \;=\; -\sum_{i=1}^{d} w_i + cS_d + (d+1). \tag{1.1}$$

**Lemma 1.3** (The instance)**.** *$w^\star = (c+1,\ 2c+1,\ \dots,\ dc+1)$ is the unique minimizer of $\Phi_P$, every point of $P$ has residual of absolute value 1 at $w^\star$, and $\Phi_P^\star := \Phi_P(w^\star) = d + 1$.*

*Proof.* At $w^\star$: $w_i^\star - ic = 1$ for $i \in [d]$, and by (1.1) the residual on $p_0$ is $-(cS_d + d) + cS_d + (d+1) = 1$. The gradient of $\frac{1}{2}\Phi_P$ at $w^\star$ is $\sum_i e_i \cdot 1 + x_0 \cdot 1 = 0$, and $\Phi_P$ is strictly convex $e_1, \dots, e_d$ span $\mathbb{R}^d$. Finally $\Phi_P(w^\star) = d \cdot 1 + 1 = d + 1$. $\square$

The next lemma is the reason the instance is hard: *every* subproblem that omits at least one of its $d+1$ constraints pays a factor of at least $d+1$.

**Lemma 1.4** (Subset lemma)**.** *Let $I \subseteq [d]$ and $b \in \{0, 1\}$ encode a set of constraints*

$$C(I, b) \;=\; \Big\{ w \in \mathbb{R}^d \;:\; w_i = ic \;\; (i \in I), \quad \textit{and} \quad \langle w, x_0\rangle = y_0 \;\; \textit{if } b = 1 \Big\},$$

*and suppose $(I, b) \neq ([d], 1)$. Then $C(I, b) \neq \emptyset$, and its minimum-norm element $\widehat{w}$ satisfies $\Phi_P(\widehat{w}) \;\geq\; (d+1)^2$.*

*Proof.* *Case* $b = 0$. The constraints fix $w_i = ic$ for $i \in I$ and leave the remaining coordinates free, so the min-norm element has $\widehat{w}_j = 0$ for $j \notin I$. By (1.1), its residual on $p_0$ is $-\sum_{i\in I} ic + cS_d + (d+1) = c\sum_{j\notin I} j + (d+1) \ \geq \ d+1$, hence $\Phi_P(\widehat{w}) \geq (d+1)^2$.

*Case* $b = 1$ Let $J := [d] \setminus I \neq \emptyset$ and $r := |J|$. The constraints are consistent: combining $w_i = ic$ ($i \in I$) with $\langle w, x_0\rangle = y_0$ and (1.1) forces exactly

$$\sum_{j\in J} w_j \ = \ c\sum_{j\in J} j \ + \ (d+1), \tag{1.2}$$

The min-norm ERM fixes $w_i = ic$ on $I$ and minimizes $\sum_{j\in J} w_j^2$ subject to (1.2), i.e. spreads the excess equally:

$$\widehat{w}_j \ = \ c\,\bar{\jmath} + \frac{d+1}{r} \quad (j \in J), \qquad \bar{\jmath} := \frac{1}{r}\sum_{j\in J} j.$$

Its loss on the missed points is, using $\sum_{j\in J}(\bar{\jmath} - j) = 0$,

$$\sum_{j\in J}\big(\widehat{w}_j - jc\big)^2 = \sum_{j\in J}\Big(c(\bar{\jmath} - j) + \tfrac{d+1}{r}\Big)^2 = c^2\sum_{j\in J}(j-\bar{\jmath})^2 \ + \ \frac{(d+1)^2}{r}.$$

If $r = 1$ this equals $(d+1)^2$. If $r \geq 2$, let $j_{\max}, j_{\min}$ be the extreme elements of $J$; then $j_{\max} - j_{\min} \geq r - 1 \geq 1$ and

$$\sum_{j\in J}(j-\bar{\jmath})^2 \ \geq \ (j_{\max} - \bar{\jmath})^2 + (\bar{\jmath} - j_{\min})^2 \ \geq \ \tfrac{1}{2}(j_{\max} - j_{\min})^2 \ \geq \ \tfrac{1}{2},$$

so

$$c^2\sum_{j\in J}(j-\bar{\jmath})^2 \geq c^2/2 > (d+1)^2$$

by the choice $c > \sqrt{2}\,(d+1)$. In both cases $\Phi_P(\widehat{w}) \geq (d+1)^2$. $\square$

**Shattering $p_0$.** Now we define the instance for general $n \geq d$. For an integer $M \geq 1$, define

$$\mu_t = 1 + \frac{t}{3M}, \qquad \lambda_t = \frac{\mu_t}{\big(\sum_{s=1}^{M}\mu_s^2\big)^{1/2}} \qquad (t = 1, \dots, M),$$

so that the $\lambda_t$ are distinct and positive, $\sum_{t=1}^{M}\lambda_t^2 = 1$, and $\max_t \lambda_t^2 \leq (4/3)^2/M \leq 2/M$. Define the $(d+M)$-point dataset

$$D_M \ = \ \{p_1, \dots, p_d\} \cup \{q_1, \dots, q_M\}, \qquad q_t := \big(\lambda_t x_0,\ \lambda_t y_0\big).$$

Since $\ell_{\lambda z}(w) = \lambda^2 \ell_z(w)$ for the squared loss, the loss landscape is unchanged: for every $w \in \mathbb{R}^d$,

$$\sum_{z\in D_M}\big(\langle w, x_z\rangle - y_z\big)^2 = \sum_{i=1}^{d}(w_i - ic)^2 + \Big(\sum_{t=1}^{M}\lambda_t^2\Big)\big(\langle w, x_0\rangle - y_0\big)^2 = \Phi_P(w). \tag{1.3}$$

In particular, by Lemma 1.3, $L^\star_{D_M} = (d+1)/(d+M) > 0$, with the same minimizer $w^\star$, and for every $w$ the ratio of its loss on $D_M$ to the optimum is $\Phi_P(w)/(d+1)$.

**Lemma 1.5.** *Let $n \ge 1$ and let $\sigma = (s_1, \dots, s_n) \in D_M^n$ be any selection (repetitions allowed). Let $\widehat{w}_\sigma = \mathcal{A}^\star\big(\frac{1}{n}\sum_{k=1}^n \ell_{s_k}\big)$. Then*

$$\frac{L_{D_M}(\widehat{w}_\sigma)}{L^\star_{D_M}} \;=\; \frac{\Phi_P(\widehat{w}_\sigma)}{d+1} \;\ge\; \frac{d+1}{\big(1+2dn/M\big)^2}\,.$$

*Proof.* Let $I = \{i \in [d] : p_i \text{ appears in } \sigma\}$, let $k_i \ge 1$ be the multiplicity of $p_i$ for $i \in I$, and let

$$\alpha \;=\; \sum_{k\,:\,s_k \in \{q_1,\dots,q_M\}} \lambda^2_{t(s_k)} \;\le\; n \cdot \max_t \lambda_t^2 \;\le\; \frac{2n}{M},$$

the total squared scale of the selected shattered copies. Using (1.1), the selected (unnormalized) objective is

$$\sum_{k=1}^{n} \ell_{s_k}(w) \;=\; \sum_{i\in I} k_i\,(w_i - ic)^2 \;+\; \alpha\,\big(\langle w, x_0\rangle - y_0\big)^2. \tag{1.4}$$

*Case 1: $I \neq [d]$ or $\alpha = 0$.* Set $b = \mathbf{1}[\alpha > 0]$; then $(I,b) \neq ([d],1)$. The constraint set $C(I,b)$ of Lemma 1.4 is nonempty, so the objective (1.4) attains minimum value 0, and its minimizer set is exactly $C(I,b)$—independently of the multiplicities $k_i$ and of which copies $q_t$ were chosen. Hence $\widehat{w}_\sigma$ is the min-norm element of $C(I,b)$ and Lemma 1.4 gives $\Phi_P(\widehat{w}_\sigma)/(d{+}1) \;\ge\; (d{+}1)^2/(d{+}1) \;=\; d+1 \;\ge\; (d+1)/(1+2dn/M)^2$.

*Case 2: $I = [d]$ and $\alpha > 0$.* Substituting $v_i = w_i - ic$ and using (1.1), the objective (1.4) becomes

$$Q(v) \;=\; \sum_{i=1}^{d} k_i v_i^2 \;+\; \alpha\Big((d+1) - \sum_{i=1}^{d} v_i\Big)^2,$$

which is strictly convex (its Hessian is $2\,\mathrm{diag}(k) + 2\alpha\mathbf{1}\mathbf{1}^\top \succ 0$), so $\widehat{w}_\sigma$ is its unique critical point and no tie-breaking is involved. It remains to evaluate the minimizer $v$. Setting $\partial Q/\partial v_i = 0$ gives, for each $i$,

$$2k_i v_i \;-\; 2\alpha\Big((d+1) - \sum_j v_j\Big) \;=\; 0, \qquad \text{i.e.} \qquad v_i \;=\; \frac{\alpha}{k_i}\Big((d+1) - \sum_j v_j\Big).$$

Take $K = \sum_i k_i^{-1}$ and sum these $d$ equations,

$$\sum_i v_i \;=\; \alpha K\Big((d+1) - \sum_i v_i\Big) \;\Longrightarrow\; (1+\alpha K)\sum_i v_i = \alpha K\,(d+1) \;\Longrightarrow\; \sum_i v_i = \frac{\alpha K\,(d+1)}{1+\alpha K}.$$

The $p_0$ residual is therefore

$$(d+1) - \sum_i v_i \;=\; (d+1)\,\frac{(1+\alpha K) - \alpha K}{1+\alpha K} \;=\; \frac{d+1}{1+\alpha K},$$

and feeding this back into $v_i = \frac{\alpha}{k_i}\big((d+1) - \sum_j v_j\big)$ yields

$$v_i = \frac{\alpha\,(d+1)}{k_i\,(1+\alpha K)}.$$

Hence, we have

$$\Phi_P(\widehat{w}_\sigma) \;=\; \sum_{i=1}^{d} v_i^2 + \Big(d+1-\sum_i v_i\Big)^2 = \frac{\alpha^2(d+1)^2\sum_i k_i^{-2}}{(1+\alpha K)^2} \;+\; \frac{(d+1)^2}{(1+\alpha K)^2}$$
$$= \frac{(d+1)^2}{(1+\alpha K)^2}\Big(1+\alpha^2\textstyle\sum_i k_i^{-2}\Big) \;\geq\; \tfrac{(d+1)^2}{(1+\alpha K)^2}.$$

Since $k_i \geq 1$ we have $K = \sum_i k_i^{-1} \leq d$, and $\alpha \leq 2n/M$, so $\Phi_P(\widehat{w}_\sigma)/(d{+}1) \;\geq\; (d{+}1)/(1{+}\alpha K)^2 \;\geq$ $(d+1)/(1+2dn/M)^2$. □

**Theorem 1.6** (Lower bound)**.** *For all $d, m \geq 1$ and every finite $n \geq d$, $\mathcal{F}(d,m,n) \geq d+1$.*

*Proof.* By Proposition 1.5, $L^\star_{D_M}(n)/L^\star_{D_M} \geq (d+1)/(1+2dn/M)^2$ with $L^\star_{D_M} > 0$. Given $\varepsilon > 0$, choosing $M \geq 2dn\big((1-\varepsilon/(d{+}1))^{-1/2}-1\big)^{-1}$ makes the right side at least $d+1-\varepsilon$. Hence $\mathcal{F}(d,1,n) \geq d+1$, padding zeros entry would lift this bound to every $m \geq 1$. □

This completes the proof of Theorem 1.1

**Remark 1.7** (Why weights help but multiplicities do not)**.** Under weighted selection, $n \geq 2d$ examples achieve ratio 1 [2, 1]. On $D_M$ this is consistent: a weighted selector may put weight $\Theta(\lambda_t^2)$ on a basis point to *match* the tiny scale of a shattered copy $q_t$, recovering $w^\star$ exactly. An unweighted selector cannot express relative weights below $1/n$, and this asymmetry is precisely what the shattering exploits.

## References


[1] Steve Hanneke, Shay Moran, Alexander Shlimovich, and Amir Yehudayoff. Data selection for ERMs. In *Proceedings of the 38th Annual Conference on Learning Theory (COLT)*, volume 291 of *PMLR*, 2025. arXiv:2504.14572.

[2] Steve Hanneke, Shay Moran, Alexander Shlimovich, and Amir Yehudayoff. Open problem: Data selection for regression tasks. In *Proceedings of the 38th Annual Conference on Learning Theory (COLT)*, volume 291 of *PMLR*, pages 1–5, 2025.